Beyond Self-Knowledge: Propagating Uncertainty Across Reasoning and Retrieval in LLMs

Anonymous submission

Abstract

Retrieval-augmented generation improves knowledge-intensive question answering, but indiscriminate retrieval can introduce irrelevant evidence and unnecessary computation. We investigate whether verbalized confidence from black-box language models can serve as an actionable signal for retrieval routing. Our method, BeyondUncertainty, first elicits a structured provisional answer and confidence estimate, then applies a model-specific threshold selected on held-out validation data and frozen before test evaluation. Low-confidence questions receive top-5 TF–IDF retrieval followed by a second answer call, whereas high-confidence questions return the provisional answer directly. We evaluate 27,000 policy instances across six QA benchmarks, three model families, and three retrieval policies. BeyondUncertainty achieves 0.483 mean token-level F1, compared with 0.467 for always retrieval and 0.401 for no retrieval, while reducing retrieved passages by 20.4% relative to always retrieval. When matched on the number of questions routed to retrieval within each dataset–model cell, it outperforms a post-hoc random allocation in 17 of 18 settings, with an average gain of 0.024 F1. Although poorly calibrated as an absolute probability, probe uncertainty modestly predicts question-level retrieval benefit (AUROC $= 0.628$). However, the additional probe increases total token usage by 28.2%, revealing a trade-off between more selective evidence acquisition and end-to-end token efficiency.

Introduction

Large language models (LLMs) can answer many factual questions from their parametric knowledge, yet they may produce fluent but unsupported responses when relevant knowledge is missing, outdated, or difficult to compose. Retrieval-augmented generation (RAG) mitigates this limitation by grounding generation in external evidence (Lewis et al. 2020). Retrieval is not uniformly beneficial, however: invoking a retriever for every question increases context length and latency, while irrelevant or conflicting passages may distract the generator and degrade answer quality (Moskvoretskii et al. 2025; Su et al. 2024; Shi et al. 2023; Yoran et al. 2024). Adaptive RAG therefore seeks to determine when external evidence is likely to be useful. Existing methods base this decision on query complexity, token-level confidence, reflection signals, internal model states, or estimated information need (Jeong et al. 2024; Asai et al. 2024; Jiang et al. 2023; Su et al. 2024). In parallel, research on LLM uncertainty examines verbalized confidence, sampling consistency, and semantic entropy as indicators of answer reliability (Kadavath et al. 2022; Xiong et al. 2024; Kuhn, Gal, and Farquhar 2023; Farquhar et al. 2024). These directions motivate a practically important question: can an uncertainty signal available through a black-box API be transferred from a structured pre-retrieval probe into a downstream retrieval decision, and can it identify which questions benefit from evidence rather than merely reducing retrieval frequency?

We investigate this question using a deliberately simple and model-agnostic controller. For each question, the LLM produces a structured provisional answer, a verbalized confidence estimate, and a concise state summary. Routing is determined solely by confidence and requires no access to hidden chain-of-thought, logits, internal activations, or model fine-tuning. A model-specific threshold, selected on held-out validation data and frozen before test evaluation, determines whether the system returns the provisional answer directly or retrieves the top-5 passages and performs a separate final-answer call. This design isolates the operational value of black-box confidence while remaining applicable to heterogeneous proprietary model families.

We address three research questions:

- **RQ1:** Does probe uncertainty predict which questions benefit from retrieval, and how does this relate to final-answer quality?
- **RQ2:** How does confidence-guided routing trade answer quality, retrieved evidence, and gateway-reported token usage?
- **RQ3:** Does confidence-guided routing outperform random allocation when both route the same number of questions to retrieval within each dataset–model cell?

Our contributions are threefold. First, we introduce a model-agnostic black-box routing interface that propagates an observable confidence signal into an external evidence-acquisition decision. Second, we conduct a paired evaluation comprising 27,000 policy instances across six single- and multi-hop QA benchmarks and three model families, retain-

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ing API and structured-output failures rather than filtering them post hoc. Third, through a route-count-matched control, we show that confidence provides question-level allocation value beyond retrieval frequency alone. Our findings also expose important limitations: verbalized confidence is poorly calibrated in absolute terms, the additional probe makes the method retrieval-saving but not token-saving, and improvements over always retrieval vary across dataset–model settings.

Related Work

Adaptive retrieval. RAG combines generation with external evidence (Lewis et al. 2020). IRCoT interleaves retrieval and reasoning (Trivedi et al. 2023); FLARE and DRAGIN trigger retrieval from token uncertainty or information need (Jiang et al. 2023; Su et al. 2024); Adaptive-RAG predicts query complexity (Jeong et al. 2024); and Self-RAG and SeaKR use reflection or self-aware knowledge signals (Asai et al. 2024; Yao et al. 2025). AdaRAGUE compares a broad family of adaptive strategies and shows that simple uncertainty signals can be competitive (Moskvoretskii et al. 2025). Our focus is narrower and complementary. Long-tail knowledge motivates retrieving only when parametric memory is unreliable (Mallen et al. 2023); RECOMP selectively suppresses unhelpful retrieved text (Xu, Shi, and Choi 2023); and robustness studies characterize how irrelevant context can reduce answer quality (Shi et al. 2023; Yoran et al. 2024). Recent work compares uncertainty-based routing, estimates retrieval need for black-box models, and uses confidence to allocate search-agent computation (Moskvoretskii et al. 2025; Lin et al. 2025; Ou et al. 2026). We isolate a single verbalized-confidence signal across heterogeneous APIs and evaluate its routing value, passage savings, and end-to-end token cost.

Uncertainty and calibration. LLM self-knowledge has been studied through answer verification and elicited probabilities (Kadavath et al. 2022). Black-box estimators include verbalized confidence, sampling agreement, and semantic dispersion (Xiong et al. 2024; Lin, Trivedi, and Sun 2023; Kuhn, Gal, and Farquhar 2023; Farquhar et al. 2024; Manakul, Liusie, and Gales 2023). Prior work also finds that generative QA probabilities and verbalized certainty can be miscalibrated (Jiang et al. 2021; Mielke et al. 2022; Tian et al. 2023; Yuan et al. 2024). Because confidence can rank predictions while remaining probabilistically miscalibrated, we separate *ordinal utility* from *absolute calibration* and report both rank association and expected calibration error (ECE) (Guo et al. 2017). We provide a controlled cross-provider study demonstrating the operational value of black-box confidence for retrieval allocation across answer quality, passage use, token cost, failures, and matched routing decisions.

Black-Box Confidence Routing

Given question q, a black-box model first returns the observable state

$$s_0 = (\hat{a}_0, c_0, d_0, m_0), \tag{1}$$

where $\hat{a}_0$ is a provisional answer, $c_0 \in [0, 1]$ is verbalized confidence, d_0 is the model's suggested action, and m_0 is a concise state summary. The prompt requests compact JSON and explicitly asks for a summary rather than hidden chain-of-thought. We define uncertainty as $u_0 = 1 - c_0$.

For model family m, the frozen router is

$$R(q, m) = \mathbb{I}[c_0 < \tau_m], \tag{2}$$

with missing or unparseable confidence conservatively mapped to retrieval. If $R = 0$, the system returns $\hat{a}_0$. If $R = 1$, a deterministic sparse retriever uses the original question as the query, returns up to five passages, and the same model receives q together with those passages in a final call:

$$P = \mathrm{TopK}(q, \mathcal{D}_j, 5), \qquad \hat{a} = \mathrm{LLM}(q, P). \tag{3}$$

Here $\mathcal{D}_j$ is the fixed corpus for dataset j. The model's suggested action d_0 is logged but does not determine routing, which keeps the decision rule reproducible and isolates the role of confidence. Figure 1 summarizes the implemented workflow. The threshold has a decision-theoretic interpretation. Let Y_0 and Y_1 denote answer utility without and with retrieval, and let C_R denote retrieval cost. A cost-sensitive controller retrieves when

$$\mathbb{E}[Y_1 - Y_0 \mid c_0, q] > \lambda C_R, \tag{4}$$

where λ encodes the value assigned to resource use. If expected retrieval benefit decreases with confidence, a threshold in c_0 provides a reduced-form approximation to this rule. We neither assume nor prove perfect monotonicity. Instead, RQ1 tests whether confidence orders observed quality, RQ2 reports quality and cost separately rather than fixing λ, and RQ3 evaluates whether the induced allocation outperforms a route-count-matched random allocation. This is a single-decision form of uncertainty propagation: an uncertainty estimate produced during provisional reasoning is consumed by a downstream retrieval component. It does *not* implement iterative uncertainty trajectories, learned routing, multi-step early stopping, or selective abstention; these remain directions for future work.

Experimental Design

Datasets and splits. We evaluate six question-answering benchmarks: Natural Questions (NQ) (Kwiatkowski et al. 2019), HotpotQA (Yang et al. 2018), 2WikiMultiHopQA (Ho et al. 2020), SQuAD (Rajpurkar et al. 2016), TriviaQA (Joshi et al. 2017), and MuSiQue (Trivedi et al. 2022). Fixed question identifiers select 500 test examples from each dataset, yielding 3,000 unique questions. Prompts and model-specific routing thresholds are selected using held-out development data and frozen before test evaluation. Test identifiers are not used for prompt design or threshold selection.

Models and decoding. We access one gateway-reported alias from each of three model families through the same OpenAI-compatible endpoint between July 8 and 11, 2026. Table 1 reports the aliases and frozen thresholds. A seed of 1337 is requested where supported, although bitwise determinism is not assumed across providers. Gemini and Claude use temperature 0 and a maximum output length of 2,000 tokens. The OpenAI endpoint rejects temperature 0; we therefore use its gateway default and a 4,000-token output limit.

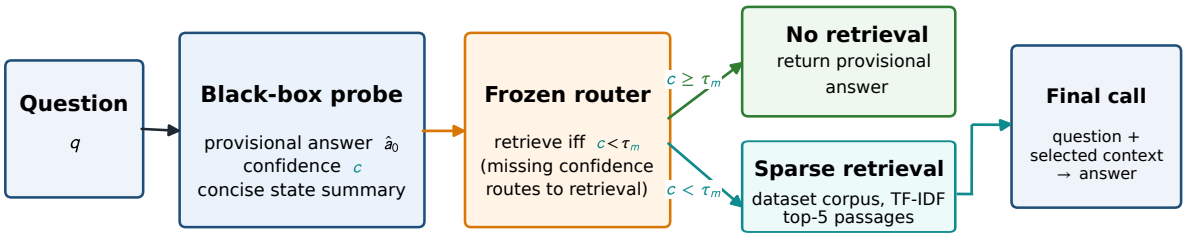


Figure 1: Implemented BeyondUncertainty pipeline. A structured black-box probe produces a provisional answer and confidence. A validation-frozen, model-specific threshold routes the question either directly to that answer or through top-5 sparse retrieval and a second answer call.

Model family	Gateway-reported alias	τ_m
OpenAI	`gpt-5.5`	1.0
Gemini	`gemini-3.5-flash`	1.0
Claude	`claude-sonnet-5`	0.9

Table 1: Gateway-reported model aliases and validation-frozen routing thresholds. Retrieval is triggered when $c_0 < \tau_m$.

Dataset	Documents	Dataset	Documents
NQ	17,687	SQuAD	14,369
HotpotQA	9,823	TriviaQA	14,696
2Wiki	6,299	MuSiQue	11,475

Table 2: Number of deduplicated title–passage documents in each controlled dataset-specific retrieval corpus.

Because none of the evaluated endpoints exposes usable token log probabilities, verbalized confidence provides the common black-box uncertainty signal.

The aliases are recorded exactly as returned by the gateway and should not be interpreted as independent verification of direct-provider model versions.

Retrieval corpus and ranking. Table 2 reports the number of deduplicated title–passage documents in each dataset-specific retrieval corpus. The operational backend is a deterministic TF–IDF retriever using English stop-word removal and word uni- and bigrams. For each benchmark, we construct a dataset-specific corpus from deduplicated title–passage pairs in the processed development and test contexts. The evaluation therefore measures routing over a controlled local corpus rather than retrieval from a production-scale Wikipedia index. All policies use the same corpus, preprocessing procedure, ranking function, and retrieval depth.

Structured probe and frozen decisions. The probe requests a compact JSON object containing a provisional answer, verbalized confidence, a suggested retrieval action, and a concise state summary of at most two sentences. The prompt prohibits Markdown fences and requests only a task-relevant summary rather than hidden chain-of-thought. The model's suggested action is recorded for analysis but does not affect routing. Model-specific thresholds are selected according to validation F1 and then frozen at 1.0 for OpenAI and Gemini and 0.9 for Claude. No prompt, threshold, retrieval depth, or routing parameter is modified after test evaluation begins. Requests are indexed by deterministic configuration hashes. Interrupted runs therefore reuse completed requests rather than issuing the same paid call again.

Policies and evaluation scale. We execute three primary policies for every question–model pair. No Retrieval performs one answer call without external context. Always Retrieve retrieves the top-5 passages before one answer call. BeyondUncertainty follows Equation 2: it performs one probe call, returns the provisional answer when retrieval is not triggered, and otherwise performs top-5 retrieval followed by a second answer call. The resulting grid contains $3{,}000 \times 3 \times 3 = 27{,}000$ executed policy rows, corresponding to 9,000 paired question–model instances. We additionally construct a deterministic route-count-matched random control and a non-deployable oracle from paired no-retrieval and always-retrieval outputs. These post-hoc controls require no additional model calls. The static policies use one generation call per example. Confidence routing uses one probe call when it does not retrieve and two calls when retrieval is triggered. Retrieved-passage counts therefore measure evidence demand, whereas total token counts capture the controller's additional inference overhead. We consequently distinguish retrieval savings from total generation cost.

Metrics and failure policy. The primary quality metric is normalized token-level F1; Exact Match (EM) is reported in the supplement. Retrieval use is measured by the number of returned passages, and token usage sums gateway-reported input and output tokens across all calls, including the probe. API and structured-output parsing failures remain in the evaluation denominator and receive zero quality scores; any raw score from malformed text is retained only for provenance. Within each of the 18 dataset–model cells, we conduct two-sided paired Wilcoxon signed-rank tests for always retrieval versus no retrieval, confidence routing versus no retrieval, and confidence routing versus always retrieval. The resulting 54 tests are jointly corrected using Holm's procedure (Holm 1979). Full per-cell results, adjusted p-values, prompts, corpus details, and failure counts are provided in the supplement.

Route-count-matched and oracle controls. Within each dataset–model cell, a stable SHA-256 ordering selects exactly the same number of questions for retrieval as Beyon-

Model	$\rho(c_0, \mathrm{F1})$	ECE_5	$\rho(u, \Delta_{\mathrm{ret}})$	AUROC
OpenAI	0.276	0.487	0.158	0.633
Gemini	0.300	0.528	0.126	0.606
Claude	0.309	0.426	0.168	0.588

Table 3: Decision utility and calibration of probe confidence. Here, $u = 1 - c_0$ and $\Delta_{\mathrm{ret}} = \mathrm{F1}_A - \mathrm{F1}_N$. AUROC measures prediction of strictly positive retrieval benefit.

dUncertainty. Selected examples use their paired always-retrieval outputs, whereas unselected examples use their paired no-retrieval outputs. The oracle selects the better of the two observed static-policy outcomes for each example. Both controls are deterministic and post hoc and require no additional model calls. The route-count-matched control tests whether confidence selects more useful questions than random routing at the same route count. Because some retrieval calls return fewer than five passages, equal route counts do not imply identical returned-passage counts. Moreover, the control does not reproduce BeyondUncertainty's structured probe or two-stage execution path; it therefore evaluates allocation rather than a fully execution-matched causal effect.

Results

RQ1: Does Confidence Predict Retrieval Benefit?

Table 3 distinguishes answer-quality prediction from retrieval-benefit prediction. Probe confidence is positively associated with final F1 across all three model families ($\rho = 0.276$–0.309), but is poorly calibrated as an absolute probability ($\mathrm{ECE}_5 = 0.426$–0.528).

We define observed retrieval benefit as

$$\Delta_{\mathrm{ret}} = \mathrm{F1}_A - \mathrm{F1}_N, \tag{5}$$

where A and N denote always retrieval and no retrieval, respectively. Across 8,837 instances with valid probe confidence, uncertainty $u = 1 - c_0$ shows a positive but modest association with retrieval benefit ($\rho = 0.154$, 95% CI $[0.131, 0.177]$) and predicts $\Delta_{\mathrm{ret}} > 0$ with AUROC 0.628 and AUPRC 0.447, against a positive-benefit prevalence of 0.354. Per-model AUROC ranges from 0.588 to 0.633.

The result is stable when failed static-policy pairs are excluded: $\rho = 0.150$, AUROC $= 0.629$, and AUPRC $= 0.448$. Mean confidence is 0.849, 0.875, and 0.655 for OpenAI, Gemini, and Claude, respectively, compared with exact-match rates of 0.364, 0.375, and 0.232. Figure 3 nevertheless shows that higher-confidence bins generally attain higher final F1. Thus, verbalized confidence is more useful as an ordinal routing signal than as a calibrated estimate of correctness or retrieval benefit.

Takeaway 1. Probe uncertainty modestly but consistently predicts question-level retrieval benefit across model families, while verbalized confidence remains poorly calibrated. It is therefore more useful as an ordinal routing signal than as a probability of correctness or retrieval benefit.

Policy	F1 ↑	Tokens / inst. ↓	Passages / inst. ↓
No retrieval	0.401	529.1	0.000
Always retrieve	0.467	1201.5	4.966
Random matched routes	0.459	1036.5	3.927
BeyondUncertainty	**0.483**	1539.8	3.952
Oracle no/always route	0.523	1054.5	3.968

Table 4: Aggregate comparison over 9,000 paired question–model instances. The random control matches the number of questions routed to retrieval within each dataset–model cell. Random and oracle results are deterministic post-hoc recombinations of static-policy outputs; their token counts exclude an independently executed routing probe.

RQ2: What Are the Quality–Efficiency Trade-offs?

Table 4 reports the aggregate policy comparison, while Figure 2 summarizes answer quality, retrieval use, and total token usage across the three executed policies. BeyondUncertainty achieves a mean F1 of 0.483, improving over no retrieval by 0.082 absolute F1 and over always retrieval by 0.016. It returns 3.952 passages per instance, compared with 4.966 for always retrieval, corresponding to a 20.4% reduction. The improvement in retrieval allocation does not translate into lower total token usage. The confidence probe increases mean token usage from 1201.5 for always retrieval to 1539.8, an increase of 28.2%. Consequently, BeyondUncertainty is *retrieval-saving* but not *token-saving* under the evaluated implementation. Table 5 shows that the aggregate improvement is heterogeneous. Confidence routing numerically outperforms always retrieval in 16 of the 18 dataset–model cells, with the two exceptions occurring for Gemini on NQ and SQuAD. However, after a single Holm correction across all 54 paired tests, only three confidence-versus-always comparisons remain significant: NQ–OpenAI and SQuAD–Claude favor confidence routing, whereas NQ–Gemini favors always retrieval.

Three patterns are particularly important. First, retrieval is not uniformly beneficial: always retrieval performs worse than no retrieval for Gemini on 2Wiki and TriviaQA and for Claude on TriviaQA. Second, the greatest selectivity appears where confidence produces a broad routing distribution. Gemini retrieves for only 19.0% of TriviaQA and 45.4% of NQ questions, while Claude retrieves for 32.2% of TriviaQA questions. Third, OpenAI routes at least 95.8% of questions to retrieval in every dataset. Its behavior is therefore close to always retrieval, leaving little opportunity for meaningful retrieval allocation. The multiplicity-corrected tests support a deliberately narrow inferential claim. Always retrieval and confidence routing each significantly outperform no retrieval in 11 of 18 cells. In contrast, confidence routing is difficult to distinguish from always retrieval after correction: two cells significantly favor confidence routing, one favors always retrieval, and the remaining 15 show no significant difference. The aggregate gain should therefore be viewed as an average improvement across heterogeneous settings rather than evidence of universal dominance.

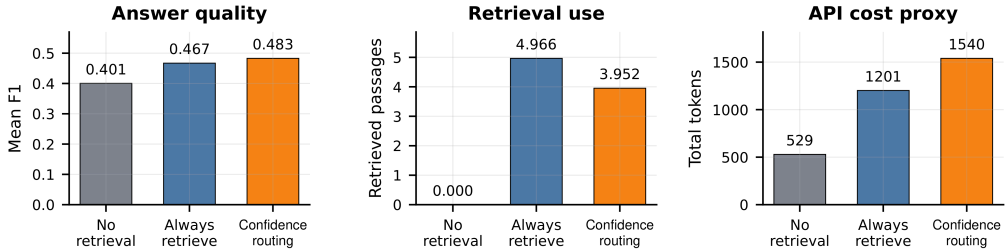


Figure 2: Aggregate answer quality and resource usage across 27,000 policy instances. Confidence routing achieves the highest mean F1 and retrieves fewer passages than always retrieval, but its additional probe increases total token usage.

Dataset	Family	F1-N	F1-A	F1-S	Δ(S–A)	Route	Dataset	Family	F1-N	F1-A	F1-S	Δ(S–A)	Route
NQ	OpenAI	0.261	0.296	**0.330**	+0.034*	0.988	SQuAD	OpenAI	0.288	0.446	**0.463**	+0.017	0.998
NQ	Gemini	0.319	**0.415**	0.368	-0.046*	0.454	SQuAD	Gemini	0.219	**0.458**	0.441	-0.016	0.842
NQ	Claude	0.193	0.252	**0.256**	+0.003	0.620	SQuAD	Claude	0.171	0.333	**0.356**	+0.024*	0.924
Hotpot	OpenAI	0.575	0.636	**0.648**	+0.013	0.998	Trivia	OpenAI	0.744	0.746	**0.763**	+0.016	0.958
Hotpot	Gemini	0.549	0.589	**0.600**	+0.011	0.584	Trivia	Gemini	0.756	0.742	**0.765**	+0.023	0.190
Hotpot	Claude	0.320	0.443	**0.459**	+0.016	0.824	Trivia	Claude	0.606	0.599	**0.635**	+0.036	0.322
2Wiki	OpenAI	0.622	0.651	**0.666**	+0.015	1.000	MuSiQue	OpenAI	0.351	0.416	**0.417**	+0.001	0.968
2Wiki	Gemini	0.530	0.505	**0.552**	+0.047	0.848	MuSiQue	Gemini	0.256	0.292	**0.330**	+0.039	0.848
2Wiki	Claude	0.320	0.377	**0.415**	+0.038	0.948	MuSiQue	Claude	0.129	0.212	**0.228**	+0.016	0.918

Table 5: Per-cell primary-policy results. N, A, and S denote no retrieval, always retrieval, and confidence routing, respectively. Route is the fraction of questions assigned to retrieval by S. A star denotes a significant S–A difference after one joint Holm correction across all 54 paired tests. Bold indicates the highest observed F1 in each dataset–model cell.

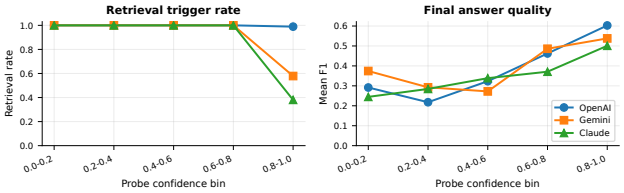


Figure 3: Retrieval rate and final F1 across probe-confidence bins. Confidence is poorly calibrated in absolute terms but generally orders questions by final answer quality.

Takeaway 2. Confidence routing achieves the highest aggregate F1 and reduces retrieved passages by 20.4% relative to always retrieval. However, its additional probe increases total token usage by 28.2%, and its cell-level advantage over always retrieval is not statistically uniform.

RQ3: Does Confidence Improve Allocation at a Matched Route Count?

The route-count-matched control retrieves for exactly the same number of questions as BeyondUncertainty within each dataset–model cell, but chooses those questions using a stable hash rather than confidence. This comparison separates the value of which questions are retrieved from the effect of route count alone. Confidence routing achieves 0.483 mean F1, compared with 0.459 for route-count-matched random routing, an average advantage of 0.024 F1. It outperforms the random allocation in 17 of the 18 cells shown in Figure 4. The only exception is MuSiQue–OpenAI, for which the difference is approximately -0.001.

The largest gains occur for MuSiQue–Gemini ($+0.043$), 2Wiki–Claude ($+0.042$), HotpotQA–Gemini ($+0.042$), and 2Wiki–Gemini ($+0.041$). These settings combine nontrivial routing rates with substantial variation in question-level retrieval benefit. Conversely, OpenAI's near-saturated routing rate leaves little difference between confidence-based and random allocations. The retrospective oracle chooses the better observed outcome between no retrieval and always retrieval for every paired example. It reaches 0.523 mean F1, leaving a 0.040 gap above BeyondUncertainty. This gap indicates meaningful headroom for improved retrieval-benefit prediction. The route-count-matched comparison nevertheless has an important limitation. It recombines outputs from the static policies and does not reproduce the confidence controller's structured probe, prompt path, or two-call execution. It therefore shows that the result is not explained by route count alone, but it does not establish a fully controlled causal effect of confidence. A directly executed randomized router with the same probe and answer-call structure would provide a stronger ablation.

Takeaway 3. At the same per-cell route count, confidence routing outperforms random allocation in 17 of 18 dataset–model cells and by 0.024 mean F1. Confidence therefore carries question-level allocation value, although the post-hoc control does not fully match the method's execution path.

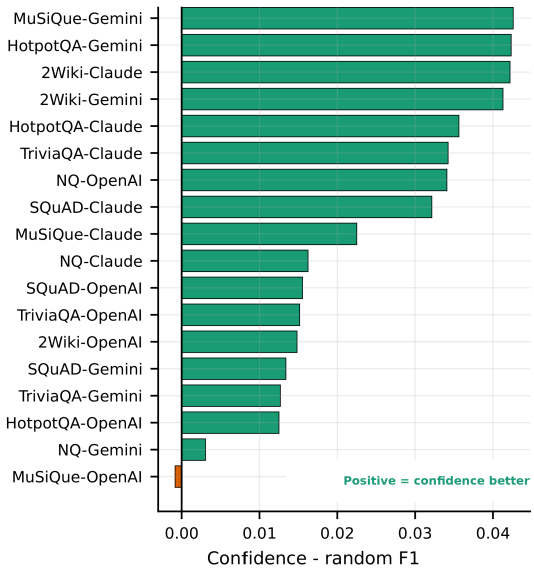


Figure 4: Per-cell F1 difference between confidence routing and random routing when both route the same number of questions to retrieval within each dataset–model cell. Positive values indicate that confidence assigns retrieval to more useful questions.

Sensitivity, Robustness, and Interpretation

Sensitivity to the routing threshold. We evaluate thresholds $\tau \in \{0, 0.3, 0.5, 0.7, 0.9, 1.0\}$ through an offline paired-output sweep. For every probe with parseable confidence, the proxy uses the corresponding always-retrieval output when $c_0 < \tau$ and the no-retrieval output otherwise. Observed probe tokens are then added to the selected branch. This procedure preserves the paired static-policy outcomes but does not issue new model calls; it should therefore be interpreted as a diagnostic sensitivity analysis rather than an independently executed policy evaluation. Among usable probe outputs, setting $\tau = 1.0$ produces proxy F1/retrieval-rate pairs of $0.538/99.2\%$ for OpenAI, $0.500/62.6\%$ for Gemini, and $0.373/99.9\%$ for Claude. The substantially different response curves demonstrate that a common numerical threshold does not induce a common retrieval budget across model families. In the executed primary-policy runs, the validation-frozen thresholds produce mean retrieval rates of 98.5% for OpenAI, 62.8% for Gemini, and 75.9% for Claude. Thus, the shared threshold of 1.0 yields almost no filtering for OpenAI but substantial filtering for Gemini. Routing behavior is determined jointly by the threshold and the model-specific confidence distribution; the numerical value of τ_m is not meaningful in isolation.

When does adaptivity help? The value of routing depends on whether retrieval has heterogeneous question-level effects. Always retrieval performs worse than no retrieval for Gemini on 2Wiki and TriviaQA and for Claude on TriviaQA, indicating that retrieved context can sometimes interfere with an otherwise correct parametric answer. Confidence routing outperforms both static policies in all three settings. By contrast, OpenAI routes at least 95.8% of questions to retrieval in every dataset. Its realized behavior is therefore close to always retrieval, leaving little opportunity for passage savings or meaningful question-level selection. The aggregate benefit of adaptivity consequently depends on both the usefulness of external evidence and the degree of variation in each model's confidence distribution.

Comparison	ΔF1	Holm sig.	Favor first	Favor second
A vs. N	+0.067	11/18	11	0
S vs. N	+0.082	11/18	11	0
S vs. A	+0.016	3/18	2	1

Table 6: Summary of 54 two-sided paired Wilcoxon tests after one joint Holm correction. N, A, and S denote no retrieval, always retrieval, and confidence routing, respectively. The aggregate difference is the mean F1 of the first policy minus that of the second.

Failure robustness and artifact completeness. The complete evaluation artifact contains all 27,000 planned primary-policy rows, forming 9,000 complete no-retrieval, always-retrieval, and confidence-routing triples. Figure 6 summarizes dataset-level coverage and the retained API and structured-output parsing failures. The artifact includes 152 API-error rows and 232 structured-output parsing failures, all of which remain in the denominator and receive zero answer-quality scores. Coverage exceeds 99% for NQ, HotpotQA, 2Wiki, SQuAD, and TriviaQA. MuSiQue has lower coverage at 94.4%, accounting for 119 API failures and 135 parsing failures. To test whether failure retention determines the main result, we additionally compute scores over successful rows only. Mean F1 changes from 0.401, 0.467, and 0.483 to 0.409, 0.472, and 0.488 for no retrieval, always retrieval, and confidence routing, respectively. The policy ordering and the rounded 0.016 confidence-versus-always gap remain unchanged. Five continuation records missing after an interrupted resume were recovered from the preceding complete artifact. Their original failure outcomes were preserved, and no answer or score was imputed. Row-level provenance, request identifiers, parsing status, and complete error counts are included in the released artifact and supplementary material.

Multiplicity-aware statistical evidence. Table 6 summarizes the paired tests after one joint Holm correction across all 54 dataset–model comparisons. Always retrieval and confidence routing each significantly outperform no retrieval in 11 of 18 cells. Confidence routing is more difficult to distinguish from always retrieval: only three comparisons survive correction, with two favoring confidence routing and one favoring always retrieval. This evidence supports a narrow interpretation. Confidence routing yields a positive aggregate improvement, but the effect is distributed heterogeneously across datasets and model families rather than appearing as a uniform advantage in every cell.

Efficiency interpretation. Figure 5 places the evaluated policies in the answer-quality–token-usage plane. Confi-

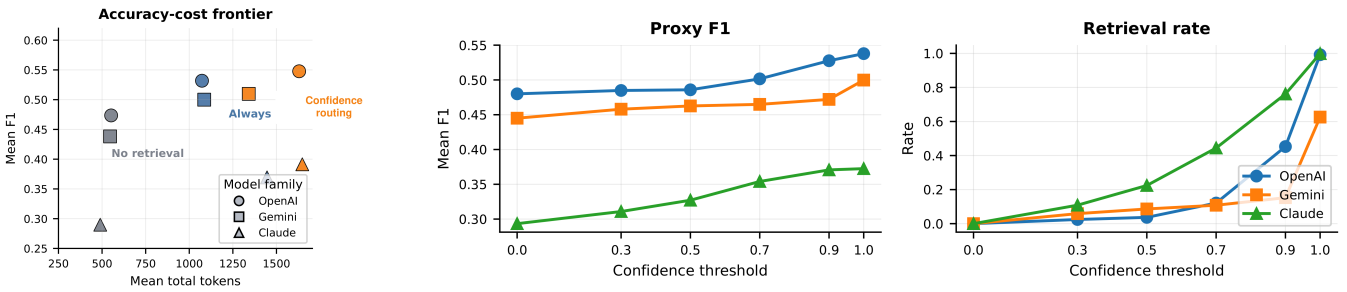


Figure 5: Sensitivity and efficiency analyses. **Left:** Mean F1 versus gateway-reported token usage for each model family and retrieval policy. **Right:** Offline sensitivity to the routing threshold, constructed by recombining paired no-retrieval and always-retrieval outputs. Increasing the threshold generally routes more questions to retrieval and improves proxy F1, but also increases passage and token usage.

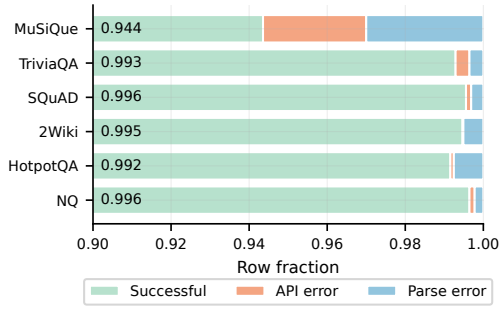


Figure 6: Coverage and retained failures across datasets. Successful, API-error, and structured-output parse-error rows partition the complete planned evaluation grid; failures remain in the scoring denominator.

dence routing occupies the highest-quality but highest-token operating point among the three executed policies: it improves mean F1 from 0.467 to 0.483 relative to always retrieval, while increasing gateway-reported token usage from 1201.5 to 1539.8 per instance. At the same time, it reduces the average number of returned passages from 4.966 to 3.952. These results reinforce the distinction between retrieval efficiency and end-to-end inference efficiency. The current controller assigns external evidence more selectively, but the additional black-box probe more than offsets those passage savings in total token usage. The post-hoc oracle should not be interpreted as a deployable point on this trade-off because it selects between outcomes after observing both static-policy results.

Takeaway 4. Routing behavior is strongly model-dependent and cannot be inferred from the numerical threshold alone. The principal findings remain unchanged when failed requests are excluded: confidence routing improves the allocation of retrieved evidence, but its additional probe prevents an end-to-end token-efficiency gain.

Discussion and Limitations

Our results distinguish the decision utility of confidence from its absolute calibration. Although verbalized confidence is overconfident, it still ranks answer quality and improves retrieval allocation relative to a route-count-matched random policy. This benefit is model-dependent and weakens when confidence values cluster near the routing threshold.

Confidence routing reduces retrieved passages by 20.4%, but the additional probe increases total token usage by 28.2%. The method therefore improves evidence allocation rather than end-to-end token efficiency. Its main limitations are the controlled TF–IDF corpus, unmatched execution paths, time-specific gateway aliases, and heterogeneous cell-level significance. Future work should explore cheaper probes, execution-path-matched controls, and frozen open-domain retrieval indexes. Additional threats to validity are provided in the supplementary material. *Overall, black-box confidence improves retrieval allocation under controlled QA conditions, but not end-to-end token efficiency.*

Ethical considerations. This study uses public QA benchmarks and involves no human participants or newly collected personal data. Because verbalized confidence is poorly calibrated and third-party gateways may raise privacy and reproducibility concerns, our claims are limited to retrieval allocation and should not be extended to safety-critical deployment without task-specific validation, monitoring, and human oversight.

Conclusion

We presented BeyondUncertainty, a black-box confidence-guided retrieval strategy for question answering. Across 27,000 policy instances, BeyondUncertainty achieves the highest aggregate F1 while reducing retrieved passages by 20.4% relative to always retrieval. Its advantage over route-count-matched random routing further indicates that verbalized confidence provides useful question-level allocation signals, despite poor absolute calibration. However, the additional probe increases total token usage, so the current method improves retrieval allocation rather than end-to-end inference efficiency. These findings motivate future work on cheaper confidence estimators, execution-path-matched controls, and open-domain retrieval settings.

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